scientific data

Check for updates

OPEN

DATA DESCRIPTOR

Chinese Food Images for Full-cycle Nutrition Analysis Towards Diabetes Management

Yuanxin Jin[1,6], Ming Li[1,7] ✉, Qinpei Zhao[2,7] ✉, Chenwei Wu[2,6], Shuang Liu[3,6], Lu Yao[3], Jessie Chen[5], Jiangfeng Li[2,7] ✉, Weixiong Rao[2], Juliang Xu[4,7] ✉ & Congrong Wang[3,7] ✉

Public food image datasets have primarily focused on recognition or segmentation, with limited resources for comprehensive nutrition analysis, particularly for Chinese cuisine which exhibits high nutritional variability due to diverse cooking methods and regional influences. To address this gap, we introduce a multimodal food image dataset designed to support full-cycle nutritional analysis, including segmentation, category recognition, and nutrient content estimation, tailored specifically to diabetic diets. This dataset is aligned with clinical data from Chinese diabetes cohorts to facilitate accurate dietary management and glucose prediction. It provides a valuable resource for advancing computational methods in meal-level nutrition assessment for diabetic populations.

Background & Summary

Tell me what you eat. I will tell you what you are. Diet plays a profound role in shaping people's health in numerous ways. A balanced diet provides the necessary nutrients, including vitamins, minerals, proteins, carbohydrates, and fats, which are essential for maintaining bodily functions and overall well-being. Dietary choices can significantly impact weight management and metabolism. Consuming a diet high in processed foods, sugars, and unhealthy fats can lead to weight gain and increase the risk of obesity, which in turn elevates the likelihood of developing chronic conditions such as type 2 diabetes (T2D), cardiovascular diseases, and certain cancers.

The rising prevalence of diabetes in China is closely linked to dietary habits among other factors. The modernization of China's society has brought about significant changes in dietary patterns, with a shift towards increased consumption of processed foods, high-calorie snacks, sugary beverages, and fatty meals. These dietary habits, coupled with reduced physical activity levels due to sedentary lifestyles, contribute to the escalating rates of obesity and type 2 diabetes in the population. Traditional Chinese diets, which were once predominantly plant-based and low in processed foods, have been gradually replaced by Westernized diets high in refined carbohydrates, saturated fats, and added sugars. As a result, there has been a surge in obesity rates and metabolic disorders, including diabetes, highlighting the critical role of dietary choices in the prevention and management of this chronic disease. Addressing these dietary habits through education, promotion of healthier food choices, and policies aimed at reducing the availability and consumption of unhealthy foods can play a crucial role in combating the diabetes epidemic in China.

However, daily dietary monitoring by individuals is tedious and inaccurate. Especially for children and adolescents who lack motivation and skills, the average error in estimating the amount of food consumed is more than 20%[1]. With the development of artificial intelligence (AI) technologies, computer-aided technical solutions have been developed to enhance and improve the efficiency and accuracy of current measurements of dietary intake[2]. Food image analysis is a method that meets the demands and can be achieved by developing algorithms in image processing and computer vision, typically using the camera on a smartphone. With the advancement of

[1]School of Electronic Information, Shanghai DianJi University, Shanghai, China. [2]School of Computer Science and Technology, Tongji University, Shanghai, China. [3]Department of Endocrinology & Metabolism, Shanghai Fourth People's Hospital, School of Medicine, Tongji University, Shanghai, China. [4]Shanghai Fengxian District Central Hospital, Shanghai, China. [5]University of Eastern Finland, Joensuu, Finland. [6]These authors contributed equally: Yuanxin Jin, Chenwei Wu, Shuang Liu. [7]These authors jointly supervised this work: Ming Li, Qinpei Zhao, Jiangfeng Li, Juliang Xu, Congrong Wang ✉e-mail: mingli@sdju.edu.cn; qinpeizhao@tongji.edu.cn; lijf@tongji.edu.cn; juliang62025@163.com; crwang@tongji.edu.cn

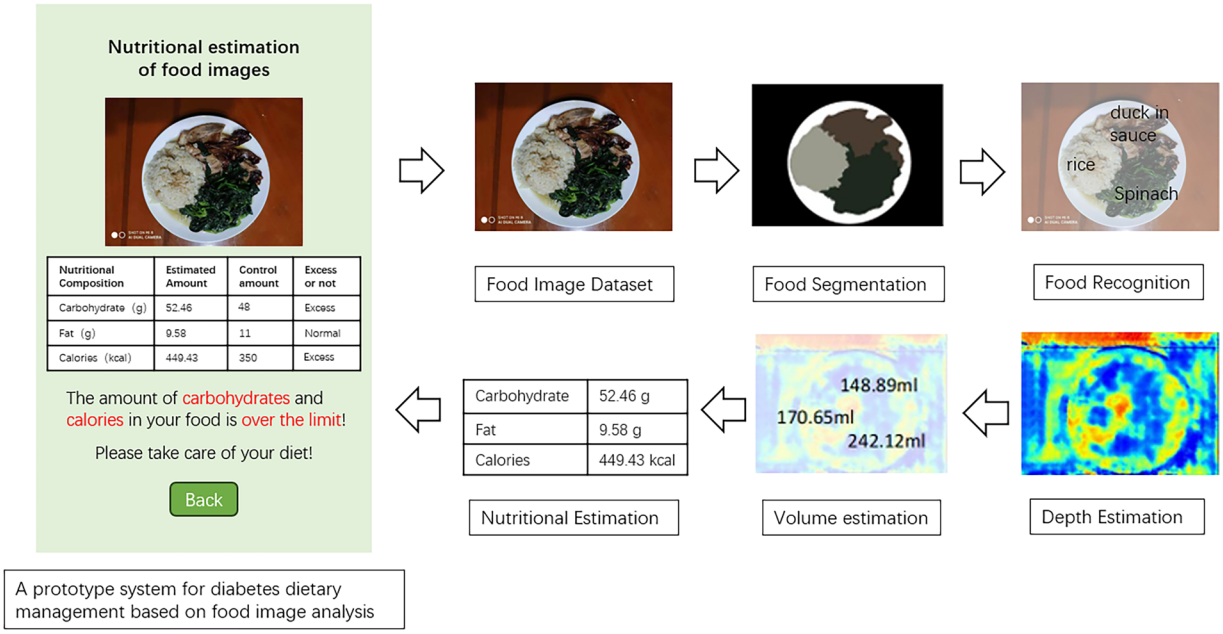


Fig. 1 A full cycle nutrition analysis on food images.

deep learning, various tasks in food image analysis have become a new interdisciplinary research field. Dietary nutrition estimation by image analysis requires construction of food database and methods on image segmentation, food recognition and volume estimation techniques. A full cycle nutrition analysis on food images is shown in Fig.1.

Many mobile apps or systems[3–5] focused on lifestyle modification, medication adherence motivation and self-management of diabetes. Support Vector Machines (SVM) was used in[6] to classify the food intake that involved a segmentation of captured images which were analyzed for feature extraction and classification. The food volume, however, was estimated using top and side views in addition to a reference image of the user's thumb. They calculated the area from the top view and multiplied the result with the depth from the side view to estimate the volume. The nutritional facts were calculated based on the weight of the food, which can be calculated using both the volume and density of individual food items. The *GoCARB* system[7,8] is a smartphone application designed to support type 1 diabetes (T1D) patients with Carbohydrates(CHO) counting by providing automatic, accurate and near real-time CHO estimation for nonpacked foods. The error of the application was below the initial goal of 20 grams, which indicated its feasibility. A dense 3D reconstruction model was presented in[9] based on two images manually captured from a mobile device camera using two different angles. Using an extrinsic camera calibration with a reference card to estimate the volume of food, the results showed that the average error rate was below 10%. In most of the applications, user intervention such as manually capturing images and placing reference objects were needed to estimate the food volume.

There are many existing food image databases with variants of food types and for different tasks. We list the public food image datasets in Table 1. There have been more than 28 datasets published in around ten years, which are for tasks such as food segmentation, recognition mainly, and nutrition or calorie estimation also. The images contain single or multiple foods and the food category varies from Asian food to western food. For Chinese food, existing literature has mostly focused on the food recognition task, e.g., the Food2K containing 2,000 food classes and 1,036,564 images completely for recognition. It is quite rare to provide a package of datasets for full-cycled tasks in nutrition analysis.

Chinese cuisine is complex due to its diverse ingredients, varied cooking methods, regional differences, intricate flavors, and significant nutritional variations based on preparation techniques. The performance of the food recognition model depends on the quality and quantity of food types and their corresponding images in the dataset. The facts make the nutrition estimation more challenging compared to the food recognition task. For example, the same ingredients can result in significantly different nutritional compositions due to varying cooking methods, such as stir-fried tomatoes with eggs (with sugar) versus tomato egg soup. In particular, the nutrition estimation for diabetic patients is quite meaningful. It can help on the accuracy of blood glucose predictions.

In this study, we introduce DFood-TJ (Diabetes-Food-Tongji), which serves as a critical visual expansion of our previously established clinical datasets, ShanghaiT1DM and ShanghaiT2DM[10]. While our earlier work laid the foundation by providing longitudinal blood glucose trajectories and electronic health records, the current DFood-TJ dataset specifically contributes high-resolution food images and manual semantic annotations. This integration allows for a more robust, multimodal approach to diabetes

Dataset	Year	Task	Food category	Total images	Food class	Type
CNFOOD-241[18]	2025	Food Recognition, Food Nutrition Estimation	Chinese food	191,811	241	Single
CAFSD[19]	2025	Food Recognition	Misc. food	21,306	239	Multi
Food2K[13]	2023	Food Recognition	Misc. food	1,036,564	2,000	Single
USDA Food and Nutrient Database for Dietary Studies 2019-2020[2]	2022	Dietary Intakes Analysis	US food	/	5,624	Single
Nutrition5k[20]	2021	Nutrition Understanding, Depth and Volume for Portion Estimation	Misc. food	5,066	/	Single, Multi
FoodSeg103[21]	2021	Food Segmentation	Western food	7,118	103	Multi
VIPER-FoodNet[22]	2021	Food Recognition	US food	14,991	82	Single, Multi
MyFood[23]	2020	Food Segmentation	Brazilian food	1,250	9	Single, Multi
ISIA Food-500[24]	2020	Food Recognition	Misc. food	399,726	500	Single
FoodX-251[25]	2019	Food Recognition	Misc. food	158,846	251	Single
ISIA Food-200[26]	2019	Food Recognition	Misc. food	197,323	200	Single
KenyanFood13[27]	2019	Food Recognition	Kenyan food	8,174	13	Single
Sushi-50[28]	2019	Food Recognition	Japanese food	3,963	50	Single
ECUSTFD[29]	2017	Food Calorie Estimation	Generic food	2,978	19	Single
UNIMIB2016[30]	2017	Food Segmentation, Food Recognition	Italian food	1,027	73	Multi
Food524DB[31]	2017	Food Recognition	Misc. food	247,636	524	Single
ChineseFoodNet[12]	2017	Food Recognition	Chinese food	185,628	208	Single
USDA National Nutrient Database for Standard Reference[32]	2016	Food and Nutrition Research	US food	/	8,789	Single
MMSPG-Food-11[33]	2016	Food Recognition	Swiss food	16,643	11	Single
UNICT-FD1200[34]	2016	Food Recognition, Food Retrieval	Italian food	4,754	8	Single
VIREO FOOD-172[35]	2016	Food Recognition, Food Retrieval	Chinese food	110,241	172	Single
Menu-match[36]	2015	Food Recognition, Food Calorie Estimation	Misc. food	646	41	Single, Multi
Food201-Segmented[37]	2015	Food Segmentation	Misc. food	12,625	201	Single, Multi
UPMC Food-101[38]	2015	Recipe Recognition	Misc. food	90,840	101	Single
Food-101[39]	2014	Food Recognition	Misc. food	10,100	101	Single
UEC FOOD-256[40,41]	2014	Food Recognition	Japanese food	31,397	256	Single, Multi
UEC FOOD-100[42,43]	2012	Food Segmentation, Food Recognition	Japanese food	9,060	100	Single, Multi
Chen[44]	2012	Food Recognition	Chinese food	5,000	50	Single

Table 1. Summary of the existing public food image datasets. Single: Single instance food images. Multi: Multi-instance food images.

dietary management, bridging the gap between clinical glucose data and real-world visual dietary intake. DFood-TJ is a core component of the broader Diabetes Data Registry and Individualized Lifestyle Intervention(DiaDRIL) study. It complements our published datasets, ShanghaiT1DM and ShanghaiT2DM, which primarily archived clinical glucose readings and dietary logs. The unique contribution of the present work is the provision of paired food imagery, enabling the transition from text-based dietary records to automated, image-based nutritional estimation for diabetic patients. Unlike our previously reported ShanghaiT1DM and ShanghaiT2DM datasets—which focus on clinical indicators and blood glucose dynamics—DFood-TJ specifically introduces a large-scale collection of food images. By providing these visual data alongside refined nutritional labels, we offer a complementary multimodal resource that enables advanced computer vision tasks, such as food segmentation and depth-based volume estimation, within a verified clinical framework.

While a majority of existing public food datasets primarily cater to classification and recognition tasks, resources supporting comprehensive, full-cycle nutritional analysis remain scarce. Among the few datasets addressing nutrient estimation, such as Nutrition5k, ECUSTFD, and the recently published CNFOOD-241, our DFood-TJ dataset distinguishes itself through three critical advantages. First, in terms of nutritional annotation, while many existing resources rely on AI-estimated values or generic database averages, the nutritional labels in DFood-TJ are derived from the actual measured weights of individual food components, meticulously verified by hospital nutrition departments to ensure the highest data integrity. Second, we provide a superior ground truth for spatial analysis through high-fidelity manual segmentation; unlike semi-automated or coarsely labeled alternatives, DFood-TJ-Seg includes over 3,900 images featuring precise manual polygon annotations that facilitate more robust model training for complex meal scenes.

Dataset	Depth Map	Nutritional Annotation	Clinical Relevance
DFood-TJ (Ours)	No	Yes (Clinical)	Yes
Nutrition5k	Yes	Yes (Detailed)	No
ECUSTFD	Yes	No (Weight/Volume only)	No
CNFOOD-241	No	No (Class labels only)	No

Table 2. Comparison of DFood-TJ with existing nutrition estimation datasets.

Finally, the most significant differentiator of DFood-TJ is its direct clinical relevance. By being explicitly synchronized with the clinical records of hospitalized diabetic cohorts from our previous ShanghaiT1DM and ShanghaiT2DM studies, this dataset provides a unique multimodal link between visual dietary intake and real-world glycemic dynamics, effectively bridging the gap between computer vision and personalized metabolic management. Consequently, as demonstrated in Table 2, DFood-TJ represents a more holistic and clinically rigorous framework for advancing automated dietary assessment in chronic disease populations.

Methods

Study population. The Diabetes Data Registry and Individualized Lifestyle Intervention registry study was initiated in 2019. Participants were recruited at two centers affiliated with Tongji University. Enrollment at Shanghai East Hospital occurred from September 2019 to March 2021; recruitment at Shanghai Fourth People's Hospital commenced later, from June 2021 to August 2025, under the same protocol.

Inclusion and Exclusion Criteria. The inclusion criteria were as follows: (1) aged 18 years or older; (2) diagnosed with diabetes according to the 1999 World Health Organization (WHO) criteria; (3) willing and able to sign the Informed Consent Form; (4) with at least 3 consecutive days of continuous glucose monitoring (CGM) records. Participants were excluded if they had a history of alcohol or drug abuse, were unable to comply with study procedures, or were deemed unsuitable for participation at the investigators' discretion.

Ethics. The study was approved by the Ethics Committees of Shanghai East Hospital and Shanghai Fourth People's Hospital, and conducted in accordance with the Declaration of Helsinki. The ethics approval numbers for the paper are as follows. The ethics approval number of Shanghai East Hospital is [2019] Body Clinical Review No. (007), and the ethics approval number of Shanghai Fourth People's Hospital is 2021071-001. Written informed consent was obtained from all participants. All data were anonymized prior to analysis to ensure patient confidentiality.

Data Collection. Participants completed a nurse-administered standardized questionnaire on demographics, family history, smoking, and alcohol intake. Clinical information, including diabetes type and duration, comorbidities, and hypoglycemic therapy, was obtained from medical records. Anthropometric measurements (height, weight, and waist circumference) were obtained using standardized methods, and body mass index (BMI) was calculated as weight (in kilograms) divided by the square of height (in meters). BMI = weight (Kg)/height(m)^2.

Continuous Glucose Monitoring and Dietary Recording. Throughout the study period, all participants were fitted with either the FreeStyle Libre system (Abbott Diabetes Care, Witney, UK) or the Guardian Connect™ CGM system (Medtronic Minimed, Inc., Northridge, CA, USA). These devices recorded interstitial glucose levels at 5-15 minute intervals, providing a high-resolution longitudinal profile of each patient's metabolic response. For precise dietary assessment, participants weighed each meal prior to consumption and documented the food type, portion size, and exact meal time. To facilitate high-fidelity image analysis, participants photographed each meal using a smartphone. These images were captured from a distance of 30-40 cm above the dishes, with meals presented without overlapping items to optimize the accuracy of automated food recognition and segmentation. In total, the DFood-TJ dataset comprises food images provided by 91 hospitalized patients. Notably, a core subset of 44 participants features dietary images that are explicitly synchronized with the clinical glucose dynamics previously reported in our ShanghaiT1DM and ShanghaiT2DM cohorts. The remaining 47 participants contribute a substantial volume of high-quality, professionally verified food imagery and nutritional labels, further enhancing the dataset's capacity for robust, full-cycle dietary assessment.

DFood-TJ. Accurate estimation of CHO and other nutrients could be beneficial in attempts to manage chronic diseases and obesity, leading to improved glycemic control and weight reduction. Methods based on photos or videos acquired using smartphone cameras have been introduced for an accurate estimation[11].

As shown in Fig. 1, a full-cycle nutrition analysis includes food images as inputs, food segmentation, food recognition and food volume estimation. The *DFood-TJ* (Table 3) is composed of food images and nutrition information of different food types. The labels of food segmentation, recognition and nutrition estimation tasks are provided. The purpose of constructing the *DFood-TJ* dataset is for a full-cycle nutrition analysis, especially for people with diabetes. Guided by our 'full-cycle' conceptual framework, we developed a multi-stage data acquisition strategy. Initially, we established *DFoodTJ-Seg* using clinical

Dataset	Description	Total images	Food class	Food type	Data Source
DFoodTJ-Nutr	an excel file for Nutrition Estimation	—	351	Single	Aqua-Calc[14], Mint Health[15]
DFoodTJ-Seg	**manual annotations for Food Segmentation**	**3,932**	**187**	**Single, Multi**	**Hospital, ChineseFoodNet**
DFoodTJ-Rec	images and labels for Food Recognition	284,933	351	Single	ChineseFoodNet, Food2K, Internet
DFoodTJ-F-Nutr	**food images with labels for Nutrition Analysis**	**3280**	**101**	**Single, Multi**	**Hospital**

Table 3. Summary of the *DFood-TJ* dataset. Single: Single instance food images. Multi: Multi-instance food images.



Nutrition Facts

Food categories and weight	Rice Braised shrimp in chili oil Fried cabbage Steamed spare ribs in black bean sauce Scallion oil winter melon	150g 50g 200g 70g 100g	Raw data, Actual measured
Carbohydrate	59.69 g		Post-processing computation
Fat	20.63 g		
Protein	22.86 g		
Fibrin	3.56 g		
Calorie	500.69 kcal		

Fig. 2 An example of the image of Chinese food with its labels of categories and nutrition.

imagery from Shanghai East Hospital and Shanghai Fourth People's Hospital. To overcome initial data scarcity, we supplemented this subset with samples from *ChineseFoodNet*[12], ensuring robust performance for image segmentation tasks. Building upon this, *DFoodTJ-Rec* was constructed using *ChineseFoodNet* as a core, further enriched by the *Food2k*[13] dataset to encompass a comprehensive variety of daily dietary patterns. Finally, through longitudinal collection efforts, we successfully curated a high-density clinical cohort to form *DFoodTJ-F-Nutr*. This specialized subset provides the ground truth labels for high-precision nutritional estimation, thereby completing the full-cycle analysis of diabetic dietary intake. As shown in Fig. 2, the dataset contains Chinese food images, food categories, corresponding weights in grams of the food, and nutrition information. The nutrition information includes the calculated carbohydrate, fat, protein, fiber, and calorie according to the weight recorded. In order to calculate the nutrition information, we constructed a food list with total of 351 food types (see the "density.xlsx" file in *DFoodTJ-Nutr*). For each food type, the list contains the density of the food (g/ml), carbohydrate to mass density (g/g), carbohydrate to volume density (g/ml), fat to mass density (g/g), fat to volume density (g/ml), protein to mass density (g/g), fiber to mass density (g/g), fiber to volume density (g/ml), calorie to mass density (kcal/g) and calorie to volume density (kcal/ml). As shown in Table 4, this is an example of nutrient estimation with rice. The density values of Chinese foods are obtained from Aqua-Calc[14] as the ground truth for volume-weight conversion, and the data for carbohydrates, fats, proteins, fiber, and calories are obtained from Mint Health[15] as the ground truth for nutrition calculation. The current data sources for the base ingredients of Mint Health are national food data, with references to the 2019 Chinese food composition tables, as well as Japanese and Korean food databases. The density of carbohydrate to volume, fat to volume, protein to volume, fiber to volume, and calorie to volume are multiplied by the mass density of the corresponding nutrient. If there is no exact match for the food category in Aqua-Calc or Mint Health, we queried the entries of its related items and averaged the corresponding density and nutrient density.

For the food image segmentation task, we constructed *DFoodTJ-Seg*, which contains 3,932 images covering 187 types of Chinese food. The manual annotation process of segmentation labels was performed manually with the support of the Labelme tool[16], which saves the annotations in a JSON file, with each mask represented by a set of polygonal points. Some examples of annotations superimposed on raw images are shown in Fig. 3. The mask of the image is generally used to block the processed image (all or part) to achieve the purpose of shielding or selecting specific elements. The process of generating the mask from the annotation is that the gray level of the pixel in the background part is set to 0, and the gray level of that in the food part is set to 1, 2, etc according to the number of food types. For example, there are two types of food in Fig. 3 (b), where the pixels in the parts of the food inside the red and green shape are set as 1 and 2, and the pixels outside are set as 0. It is similar for Fig. 3 (a) and (c). The *DFoodTJ-Seg* can be used to construct a deep-learning based segmentation model. It can also be used to perform the evaluation on the segmentation model.

After the food segments are obtained from the segmentation task, food recognition is the following task. There have been many open data sets for the recognition task (see Table 1). In our case, the *DFoodTJ-Rec* dataset contains 284,933 images of 351 Chinese foods, some examples were shown in Fig. 4. The *DFoodTJ-Rec* is constructed based on the *ChineseFoodNet* dataset, by gradually adding more images of more food types from other resources such as *Food2K*, image search engines. Although the *Food2K* dataset contains a larger number of images and food categories (2, 000), most of its images come from restaurants rather than typical home-cooked meals.

food type	Nutrients	Mass Density (g/g)	Volumetric Density (g/ml)
rice	Overall Density	0.7300	
	Carbohydrate	0.2590	0.1891
	Fat	0.0030	0.0022
	Protein	0.0260	0.0190
	Cellulose	0.0030	0.0022
	Caloric Content	1.1600	0.8468

Table 4. Mass density (data from Aqua-Calc[14]) and volumetric density (data from Mint Health[15]) of various nutrients of rice.

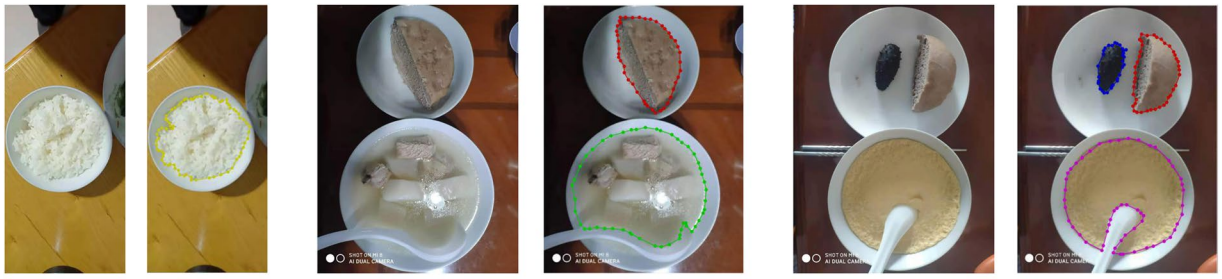


Fig. 3 Original food images and annotations on different food types. The shapes with dots and lines around the food are the manual annotations, which could be used as ground-truth labels for an image segmentation task.

Based on the nutritional information provided in the *DFoodTJ-Nutr*, nutrient content can be estimated once the food type is identified and its volume or weight has been determined. The weight of the food item may be obtained either through manual recording or estimated via computer vision techniques.

In *DFoodTJ-F-Nutr*, we provide 3280 images of 101 food types (examples see Fig. 5). Part of them were collected from the diabetes patients under doctor's advices. These images were captured using smartphone cameras, and the patient's food for each meal was weighed to record their dietary intake. During the image acquisition process, all food items were required to be placed together on a large plate. Two photographs were taken: one from a directly overhead angle and another at a 45-degree oblique angle. Each food category was annotated, and the total intake mass was provided.

Data Records

The *DFood-TJ* comprises four folders named "DFoodTJ-F-Nutr", "DFoodTJ-Nutr", "DFoodTJ-Rec", and "DFoodTJ-Seg". Table 3 shows the summary of the *DFood-TJ* dataset. The folder structure of the *DFood-TJ* is shown in Fig. 6.

The folder "DFoodTJ-F-Nutr" comprises two sub-folders: "img" and "truth". The "img" sub-folder contains 3,280 images of Chinese food, all captured from the daily meals of 91 hospitalized diabetic patients.

The "truth" sub-folder includes two files: "truth.xls" and "truth_raw.xlsx". The "truth.xls" file contains image identifiers, food identifiers, and corresponding nutrient mass values including carbohydrates, fats, proteins, fiber, and calories. Here, the image identifiers correspond to the numbered images in the "img" folder; the food identifiers represent a composite of all food types present in each image; and the nutrient sections provide the actual mass values of each nutrient per image. The "truth_raw.xlsx" file contains more detailed entries, including image identifiers, patient identifiers, date of capture, Chinese food names, food identifiers, food weights, as well as computed values for carbohydrates, fats, proteins, fiber, and calories. Among these, the image identifiers correspond to those in the "img" folder; the dates indicate when each image was taken; the food weights reflect the actual measured values during patient meals; and the nutrient data are calculated based on these weights.

The "DFoodTJ-Nutr" contains a file named 'density.xlsx', which includes the food types in Chinese, the density of Chinese food, as well as the mass density and volume density of carbohydrates, fats, proteins, fiber, and calories. These densities are obtained from Aqua-Calc[14] and Mint Health[15]. The translation of the food types is also provided in the file.

The "DFoodTJ-Rec" contains the following folders:

- The train folder contains training images, which are placed in 351 folders according to food categories;
- The val folder contains validating images, which are placed in 351 folders according to food categories;
- The test folder contains testing images, which are placed in 351 folders according to food categories;

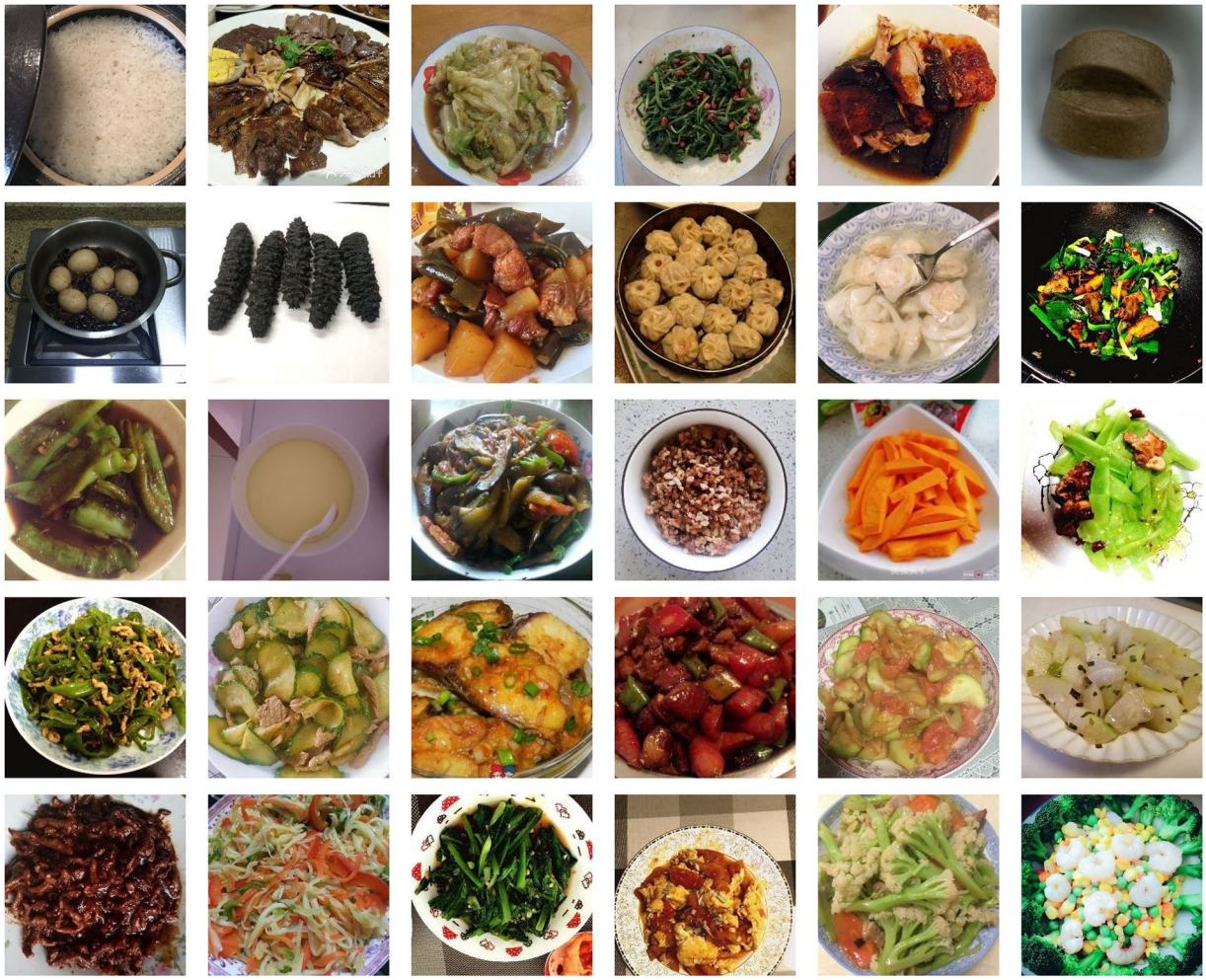


Fig. 4 Examples of the images in the recognition dataset.

- The truth_list folder contains three files named train_list.txt file, val_list.txt file and test_list.txt file. The train_list.txt file includes the number of sub-folder in the train folder, the number of the picture, and the food category number of each training image; the val_list.txt file includes the number of sub-folder in the val folder, the number of the picture, and the food category number of each validating image; the test_list.txt file includes the number of sub-folder in the test folder, the number of the picture, and the food category number of each testing image.

The "DFoodTJ-Seg" comprises three folders named "generating_program", "img" and "maskrcnn_groundtruth", The folder "generating_program" contains the programs used to generate masks from the original input images. The folders "img" and "maskrcnn_groundtruth" store 3,932 original input images and the annotated image masks, respectively.

Technical Validation

The DFoodTJ-F-Nutr dataset was collected at Shanghai Fourth People's Hospital with the consent and cooperation of diabetic patients, who require strict dietary management. During hospitalization, the hospital's nutrition department captured images and recorded detailed dietary intake for each meal. All food images and their associated weight data were manually reviewed and verified by a team of endocrinologists to ensure accuracy and reliability. Images that did not meet quality standards were excluded from the dataset. This rigorous quality control protocol was essential to maintain dataset integrity and suitability for subsequent research. We calculated the mean, standard deviation (SD), and mean absolute deviation (MAD) of the nutrients intake for patients during breakfast, lunch, and dinner respectively, as shown in Table 5. Nutritional intake metrics including the mean, standard deviation (SD), and mean absolute deviation (MAD) per serving, were calculated for patients stratified by BMI, as detailed in Table 6. Figure 7 illustrates the 20 most frequent food items within the *DFoodTJ-F-Nutr* dataset, along with their respective occurrence frequencies and percentage proportions. The distribution of daily nutrient intake across the 91 patient cohort is presented in Fig. 8. Each subplot corresponds to a specific nutrient (carbohydrate, fat, protein, fiber, and calories), utilizing boxplots



Fig. 5 Examples of the food images in dataset *DFood-TJ*.

```
DFoodTJ
├── DFoodTJ-F-Nutr
│   ├── img
│   └── truth
│       ├── truth.xls
│       └── truth_raw.xlsx
├── DFoodTJ-Nutr
│   └── density.xlsx
├── DFoodTJ-Rec
│   ├── test
│   ├── train
│   ├── truth_list
│   └── val
└── DFoodTJ-Seg
    ├── generating_program
    ├── img
    └── maskrcnn_groundtruth
```

Fig. 6 The structure of the *DFood-TJ* dataset.

Meal	Diversity	Nutrition	Mean	SD	MAD
Breakfast	6 Types	Carbohydrate	63.80 g	10.36 g	6.60 g
		Fat	10.03 g	1.94 g	0.75 g
		Protein	21.54 g	2.56 g	1.46 g
		Fiber	3.97 g	0.80 g	0.42 g
		Calorie	621.57 kcal	338.96 kcal	287.18 kcal
Lunch	58 Types	Carbohydrate	72.05 g	17.93 g	13.94 g
		Fat	26.27 g	19.19 g	14.09 g
		Protein	25.19 g	8.10 g	6.39 g
		Fiber	4.86 g	2.14 g	1.56 g
		Calorie	584.19 kcal	196.50 kcal	150.76 kcal
Dinner	54 Types	Carbohydrate	71.27 g	17.31 g	14.33 g
		Fat	20.74 g	13.33 g	9.52 g
		Protein	23.38 g	8.29 g	6.77 g
		Fiber	4.91 g	1.53 g	1.16 g
		Calorie	521.96 kcal	152.09 kcal	113.03 kcal

Table 5. The mean, standard deviation and mean absolute deviation of carbohydrate, fat, protein, fiber, and calories in the *DFoodTJ-F-Nutr* dataset, split by meal.

BMI	Nutrition	Mean	SD	MAD
<28	Carbohydrate	14.52 g	20.22 g	14.90 g
	Fat	8.24 g	11.80 g	7.32 g
	Protein	8.33 g	6.28 g	4.55 g
	Fiber	1.08 g	0.98 g	0.77 g
	Calorie	167.31 kcal	147.78 kcal	105.45 kcal
>=28	Carbohydrate	16.02 g	21.84 g	17.11 g
	Fat	6.55 g	9.73 g	5.36 g
	Protein	7.98 g	6.05 g	4.38 g
	Fiber	1.18 g	1.08 g	0.83 g
	Calorie	146.32 kcal	114.24 kcal	90.46 kcal

Table 6. The mean, standard deviation and mean absolute deviation of carbohydrate, fat, protein, fiber, and calories in the *DFoodTJ-F-Nutr* dataset, split by BMI.

to visualize data distribution through five key statistics. The interquartile range (IQR) is represented by the blue box, the median by a red line, and the mean by a green diamond. Individual patient means are plotted as blue dots, with a random jitter applied to prevent overlap and highlight individual variations. To validate the reliability of the *DFoodTJ-F-Nutr* dataset for nutritional estimation, we employed a ResNet101 architecture pre-trained on ImageNet. The model was trained using the Adam optimizer ($lr = 0.0003$, $batch_size = 16$) for 80 epochs, with a 8:2 training-to-validation split. Data augmentation techniques, including random cropping and horizontal flipping, were applied during training. The resulting per-plate estimation errors on the validation set were 21.14% for carbohydrates, 54.88% for fat, 40.56% for protein, 39.36% for fiber, and 33.01% for total calories.

To evaluate the reliability of the *DFoodTJ-Rec* dataset, a ResNet50 architecture pre-trained on ImageNet was employed. The model was trained using the Adam optimizer ($lr = 0.0005$, $batch_size = 256$) for 10 epochs. We maintained the original dataset configuration with a train:val:test split ratio of 8:1:1. Standard data augmentation techniques, including random cropping and horizontal flipping, were applied during the training phase. The resulting model achieved a top-1 accuracy of 76.57% and a top-5 accuracy of 94.35% on the test set, demonstrating the robust discriminative power of the dataset labels.

The *DFoodTJ-Seg* dataset was manually annotated and constructed. To verify its reliability, we re-annotated it by training a U-Net network structure and calculated the intersection over union (IoU) and Dice coefficient between the U-Net re-annotation results and the manual annotations to assess the annotation quality of the *DFoodTJ-Seg* dataset. We utilized a standard hold-out validation set. Specifically, we took 80% of the *DFoodTJ-Seg* dataset images as the training dataset and 20% as the validation dataset, with the epoch set to 20. The final results showed an IoU of 0.874 and a Dice coefficient of 0.924. The experimental results indicate that the *DFoodTJ-Seg* dataset is highly reliable for the task of food segmentation.

Data availability

The dataset has been made publicly available via Science Databank[17] and can be accessed at https://doi.org/10.57760/sciencedb.28012.

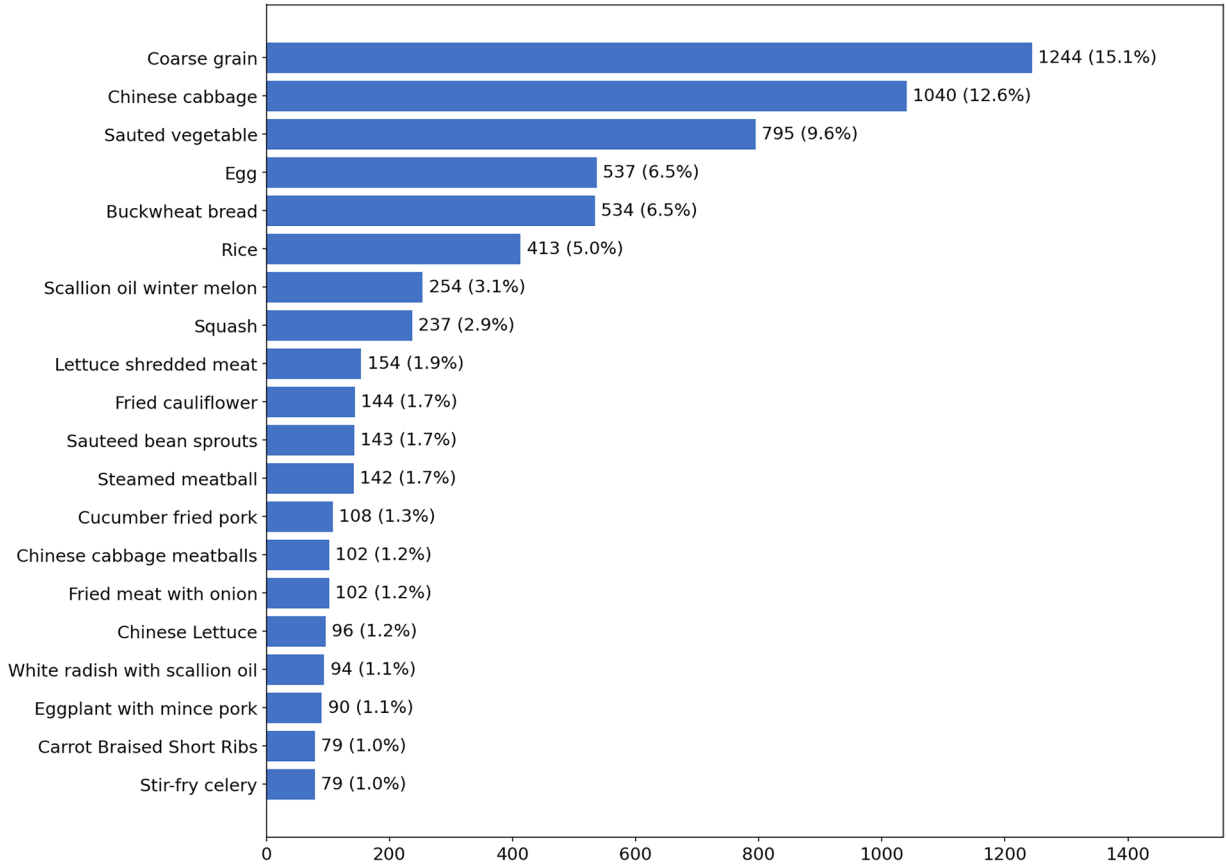


Fig. 7 The 20 most common food in the *DFoodTJ-F-Nutr*.

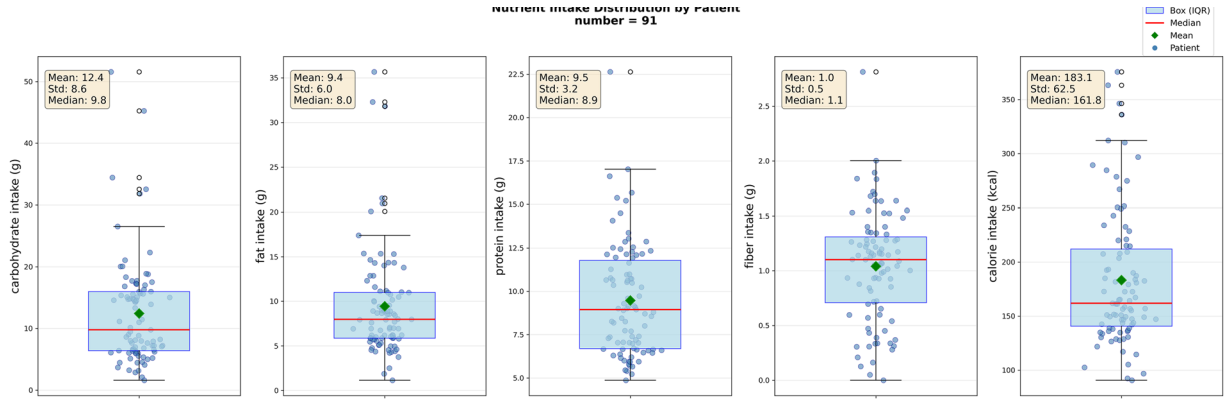


Fig. 8 Nutrient intake distribution by patient.

Code availability

The scripts used in this paper can be accessed in the Science Databank[17]. They include an image recognition script named Rec.py, a nutrition estimation script named Nutr.py, and an image segmentation script named Seg.py. The script can be executed with Python 3.6 and allows for reproducibility and code reuse.

Received: 19 September 2025; Accepted: 22 April 2026;
Published online: 29 April 2026

References

1. Hernández, T. *et al*. Portion size estimation and expectation of accuracy. *J. Food Compos. Anal.* **19**, S14–S21 (2006).
2. Konstantakopoulos, F. S., Georga, E. I. & Fotiadis, D. I. A review of image-based food recognition and volume estimation artificial intelligence systems. *IEEE Rev. Biomed. Eng.* **17**, 136–152 (2024).
3. Pouladzadeh, P., Shirmohammadi, S. & Yassine, A. You are what you eat: So measure what you eat! *IEEE Instrum. Meas. Mag.* **19**, 9–15 (2016).

4. Subhi, M. A., Ali, S. H. M., Ismail, A. G. & Othman, M. Food volume estimation based on stereo image analysis. *IEEE Instrum. Meas. Mag.* **21**, 36–43 (2018).
5. Doupis, J. *et al*. Smartphone-based technology in diabetes management. *Diabetes Ther.* **11**, 607–619 (2020).
6. Pouladzadeh, P., Shirmohammadi, S. & Al-Maghrabi, R. Measuring calorie and nutrition from food image. *IEEE Trans. Instrum. Meas.* **63**, 1947–1956 (2014).
7. Anthimopoulos, M. *et al*. Computer vision-based carbohydrate estimation for type 1 patients with diabetes using smartphones. *J. Diabetes Sci. Technol.* **9**, 507–515 (2015).
8. Rhyner, D. *et al*. Carbohydrate estimation by a mobile phone-based system versus self-estimations of individuals with type 1 diabetes mellitus: a comparative study. *J. Med. Internet Res.* **18**, e101 (2016).
9. Dehais, J., Anthimopoulos, M., Shevchik, S. & Mougiakakou, S. Two-view 3D reconstruction for food volume estimation. *IEEE Trans. Multimedia* **19**, 1090–1099 (2016).
10. Zhao, Q. *et al*. Chinese diabetes datasets for data-driven machine learning. *Sci. Data* **10**, 35 (2023).
11. Vasiloglou, M. F. *et al*. Multimedia Data-Based Mobile Applications for Dietary Assessment. *J. Diabetes Sci. Technol.* **17**, 1056–1065 (2023).
12. Chen, X., Zhu, Y., Zhou, H., Diao, L.& Wang, D. Chinesefoodnet: A large-scale image dataset for chinese food recognition. Preprint at https://arxiv.org/abs/1705.02743 (2017).
13. Min, W. *et al*. Large scale visual food recognition. *IEEE Trans. Pattern Anal. Mach. Intell*. (2023).
14. Aqua-Calc Team. *Food Volume-to-Weight Calculation Dataset*. https://www.aqua-calc.com/calculate/food-volume-to-weight (2025).
15. Mint Health Research Team. *Food Nutrition Information Dataset*, [2025-8-15]. https://www.boohee.com/food (2025).
16. Wkentaro. *Labelme: Image Polygonal Annotation Tool*.https://github.com/wkentaro/labelme (2024).
17. Jin, Y. *et al*. Chinese Food Images for Full-cycle Nutrition Analysis Towards Diabetes Management[DS/OL]. *V3. Science Data Bank* [2025-09-18], https://doi.org/10.57760/sciencedb.28012 (2025).
18. Feng, Y. *et al*. Large-scale image classification and nutrient estimation for Chinese dishes. *J. Agric. Food Res.* **19**, 101733 (2025).
19. Karabay, A., Varol, H. A. & Chan, M. Y. Improved food image recognition by leveraging deep learning and data-driven methods with an application to Central Asian Food Scene. *Sci. Rep.* **15**, 14043 (2025).
20. Thames, Q. *et al*. Nutrition5k: Towards automatic nutritional understanding of generic food. In *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit*. 8903-8911 (IEEE/CVF, 2021).
21. Wu, X. *et al*. A large-scale benchmark for food image segmentation. In *Proc. 29th ACM Int. Conf. Multimedia* 506–515 (ACM, 2021).
22. Mao, R. *et al*. Visual aware hierarchy based food recognition. In *Int. Conf. Pattern Recognit*. 571–598 (Springer, 2021).
23. Freitas, C. N. C., Cordeiro, F. R. & Macario, V. MyFood: A food segmentation and classification system to aid nutritional monitoring. In *2020 33rd SIBGRAPI Conf. Graphics, Patterns and Images* 234–239 (IEEE, 2020).
24. Min, W. *et al*. ISIA Food-500: A dataset for large-scale food recognition via stacked global-local attention network. In *Proc. 28th ACM Int. Conf. Multimedia* 393–401 (ACM, 2020).
25. Kaur, P., Sikka, K., Wang, W., Belongie, S. & Divakaran, A. Foodx-251: a dataset for fine-grained food classification. Preprint at https://arxiv.org/abs/1907.06167 (2019).
26. Min, W. *et al*. Ingredient-guided cascaded multi-attention network for food recognition. In *Proc. 27th ACM Int. Conf. Multimedia* 1331–1339 (ACM, 2019).
27. Jalal, M. *et al*. Scraping social media photos posted in Kenya and elsewhere to detect and analyze food types. In *Proc. 5th Int. Workshop on Multimedia Assisted Dietary Management* 50–59 (ACM, 2019).
28. Qiu, J., Lo, F. P.-W., Sun, Y., Wang, S. & Lo, B. Mining discriminative food regions for accurate food recognition. Preprint at https://arxiv.org/abs/2207.03692 (2022).
29. Liang, Y. & Li, J. Computer vision-based food calorie estimation: dataset, method, and experiment. Preprint at https://arxiv.org/abs/1705.07632 (2017).
30. Ciocca, G., Napoletano, P. & Schettini, R. Food recognition: a new dataset, experiments, and results. *IEEE J. Biomed. Health Inf.* **21**, 588–598 (2016).
31. Ciocca, G., Napoletano, P. & Schettini, R. Learning CNN-based features for retrieval of food images. In *New Trends in Image Analysis and Processing - ICIAP* 426–434 (Springer, 2017).
32. U.S. Department of Agriculture, Agricultural Research Service, Nutrient Data Laboratory. USDA National Nutrient Database for Standard Reference, Release 28 (Slightly Revised). *National Agricultural Library Data Repository* https://fdc.nal.usda.gov/ (2016).
33. Singla, A., Yuan, L. & Ebrahimi, T. Food/non-food image classification and food categorization using pre-trained GoogLeNet model. In *Proc. 2nd Int. Workshop on Multimedia Assisted Dietary Management* 3–11 (ACM, 2016).
34. Farinella, G. M. *et al*. Retrieval and classification of food images. *Comput. Biol. Med.* **77**, 23–39 (2016).
35. Chen, J. & Ngo, C.-W. Deep-based ingredient recognition for cooking recipe retrieval. In *Proc. 24th ACM Int. Conf. Multimedia* 32–41 (ACM, 2016).
36. Beijbom, O. *et al*. Menu-match: Restaurant-specific food logging from images. In *Proc. IEEE Winter Conf. Appl. Comput. Vis*. 844–851 (IEEE, 2015).
37. Meyers, A. *et al*. Im2Calories: towards an automated mobile vision food diary. In *Proc. IEEE Int. Conf. Comput. Vis*. 1233–1241 (IEEE, 2015).
38. Wang, X. *et al*. Recipe recognition with large multimodal food dataset. In *2015 IEEE Int. Conf. Multimedia & Expo Workshops* 1–6 (IEEE, 2015).
39. Bossard, L., Guillaumin, M. & Van Gool, L. Food-101—mining discriminative components with random forests. In *Computer Vision - ECCV 2014* 446-461 (Springer, 2014).
40. Kawano, Y. & Yanai, K. Automatic expansion of a food image dataset leveraging existing categories with domain adaptation. In *Computer Vision - ECCV 2014 Workshops* 3–17 (Springer, 2015).
41. Kawano, Y. & Yanai, K. FoodCam-256: A large-scale real-time mobile food recognition system employing high-dimensional features and compression of classifier weights. In *Proc. 22nd ACM Int. Conf. Multimedia* 761–762 (ACM, 2014).
42. Kawano, Y. & Yanai, K. Real-time mobile food recognition system. In *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops* 1–7 (IEEE, 2013).
43. Matsuda, Y., Hoashi, H. & Yanai, K. Recognition of multiple-food images by detecting candidate regions. In *2012 IEEE Int. Conf. Multimedia and Expo* 25–30 (IEEE, 2012).
44. Chen, M.-Y. *et al*. Automatic Chinese food identification and quantity estimation. In *SIGGRAPH Asia 2012 Technical Briefs* 1–4 (ACM, 2012).

Acknowledgements

This work was supported by The Noncommunicable Chronic Diseases-National Science and Technology Major Project (2023ZD0509206), the Tongji University Medical-Engineering Interdisciplinary Fund (No.2025-0585-YB-03), the Shanghai Gaofeng Project for University Academic Program Development, and the Shanghai Educational Science Research General Project [No. C2025084].

Author contributions

Qinpei Zhao, Ming Li and Congrong Wang had the initial idea for this study. Yuanxin Jin, Chenwei Wu, Shuang Liu and Lu Yao established the datasets, i.e., *DFood-TJ*. Jessie Chen, Jiangfeng Li and Weixiong Rao verified the data. Yuanxin Jin and Chenwei Wu designed and performed the technical validation. Yuanxin Jin, Chenwei Wu and Shuang Liu drafted the paper. Ming Li, Qinpei Zhao, Jiangfeng Li, Juliang Xu and Congrong Wang jointly supervised the work. All authors participated in verifying the data and revising the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to M.L., Q.Z., J.L., J.X. or C.W.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

© The Author(s) 2026